Tallinn University of Technology

Department of Mechanical and Industrial Engineering

Homework II **MTM0070 Materials Engineering**	
Title of the homework: Casting, forming and machining	
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Number 4

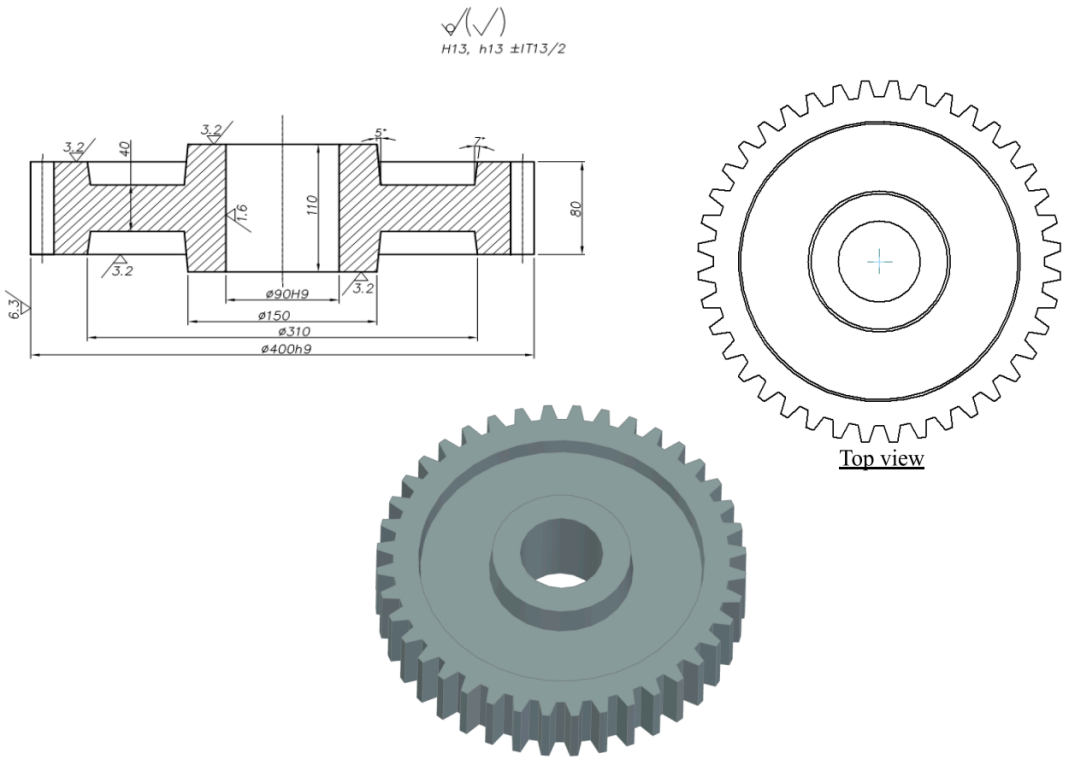


Fig. 1 Forging drawing (according to your study code last number)

Hot forging

Table 1 Forging part drawing no, material, forging method and press type (fill according to your student code last number)

Study code last number	Number of the drawing	Material	Forging method	Forging press
3	4	Steel	Closed die with flash	Form Press

1. Analyse the forging method for advantages and disadvantages. Determine the forging temperature according to the forged material. Fill in the Table 2. (12 p)

Table 2 Analysis of forging method

Material	Forging method	Forging method advantages	Forging method disadvantages	Forging temperature, (°C)
Steel	Closed die with flash	High Strength, reproducibility	High die cost, machining needed	850-1150

2. Forging sketch – draw the forging sketch. Show the parting line, forging draft angles, machining allowance. (10 p)

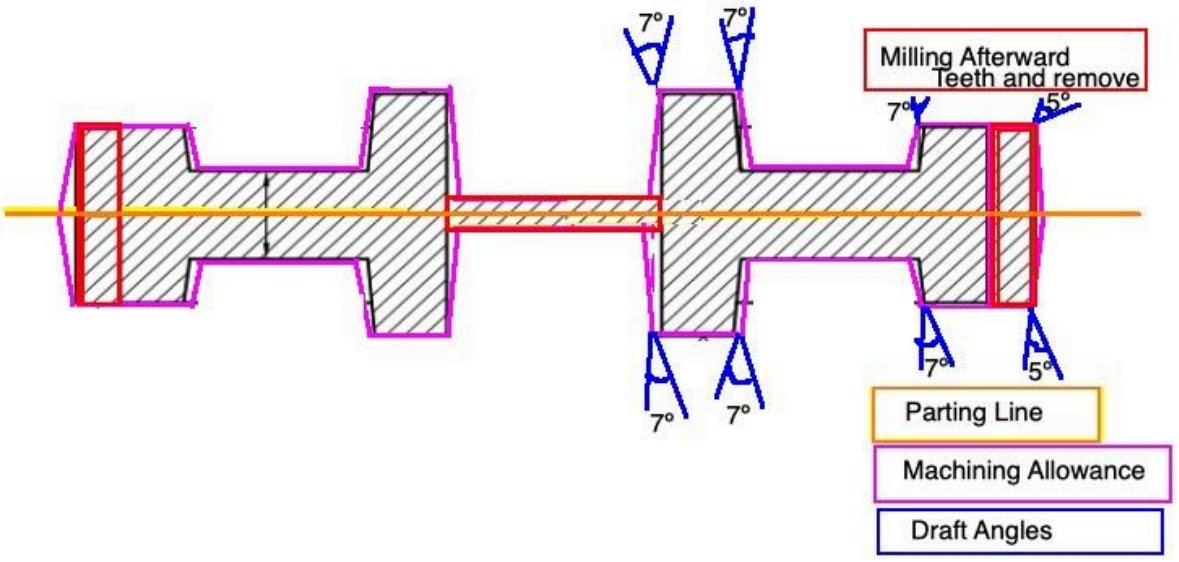


Fig. 2 Forging sketch.

3. Forging scheme - Make the drawing of the forging scheme – forging and the forging tool. Show the forging scheme and flash type. (10 p)

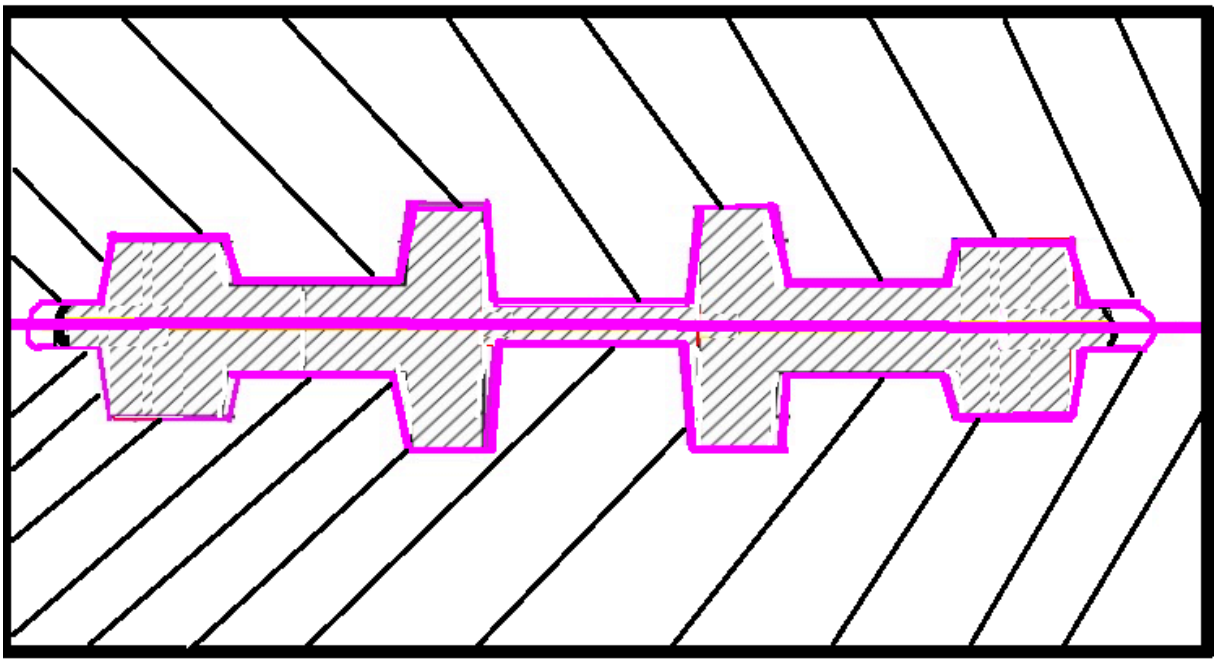


Fig. 3 Example of the forging scheme: Closed die with flash

Sand casting

Table 3 Material and position of the casting (fill according to your student code last number)

Study code last number	Number of the drawing	Material	Position of casting
3	4	Carbon Steel	Vertical

1. Casting method analysis - Analyse the casting method for advantages and disadvantages, select casting temperature. Fill in the Table 4 (12 p)

Table 4 Analysis of casting method and material

Material	Shrinkage, %	Casting method	Casting method advantages	Casting method disadvantages	Casting tempera-ture, (°C)
Carbon Steel	2.6	Sand Casting	Any Shape, Low Cost	Rough Surface, wider tolerance	1350-1450

2. Casting sketch - Mark the parting surface, draft angles, machining allowance on the part drawing (10 p)

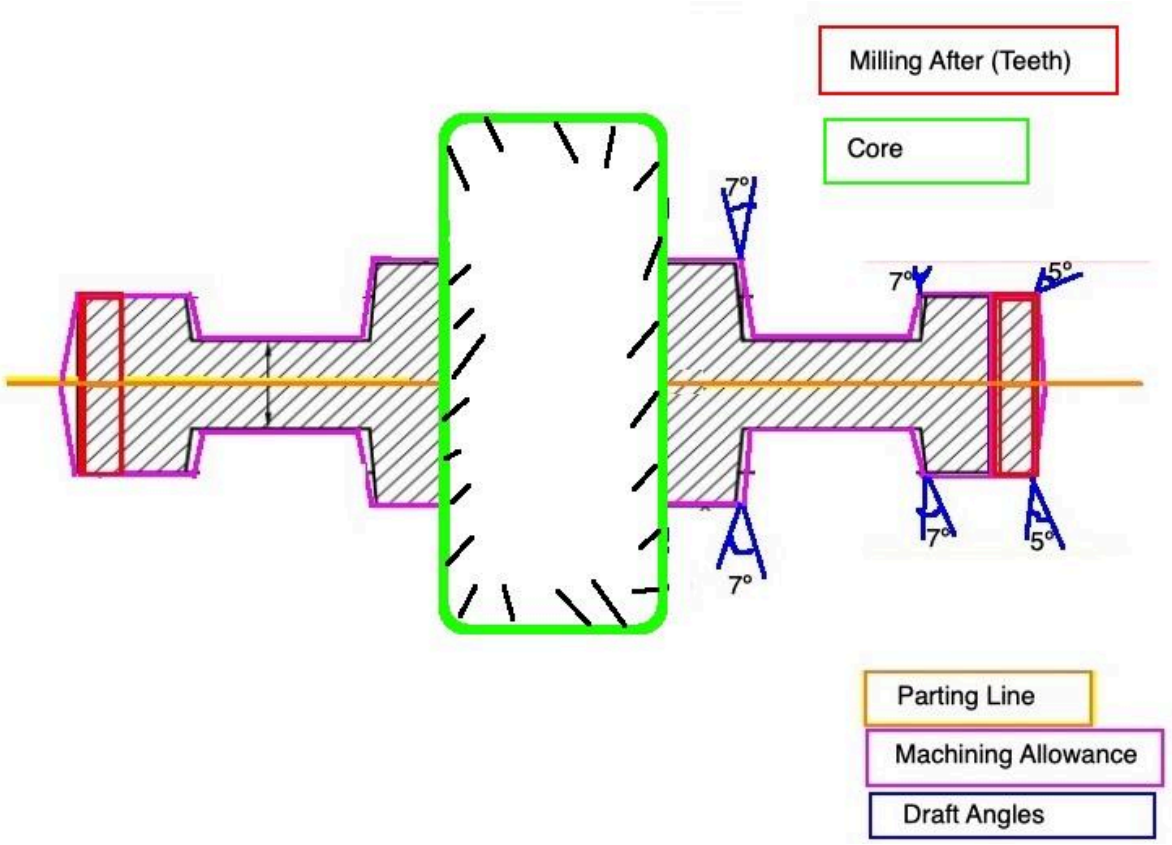


Fig. 4 Cast part sketch along with core, draft angles.

3. Pattern sketch - Draw the pattern (split or not). Show the parting surface, shrinkage allowance, lower and upper part. Describe the type of the pattern (split or one piece). (10 p)

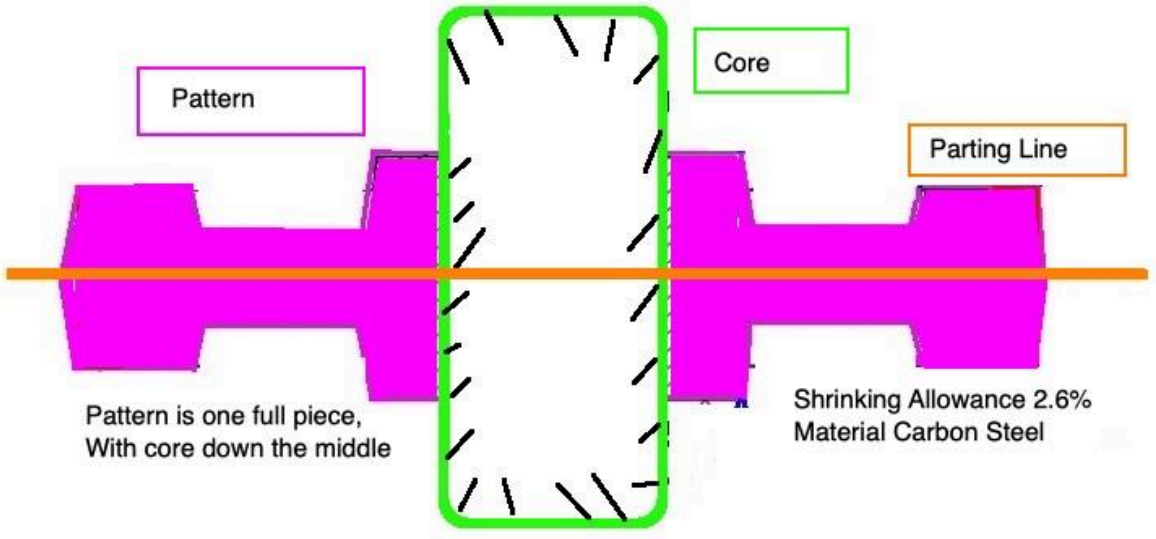


Fig. 5 Casting pattern sketch along with cast material and shrinkage allowance

4. Casting mould sketch - Draw mould along with mould cavity, flask, core and gating system. Designate the mould components. (10 p)

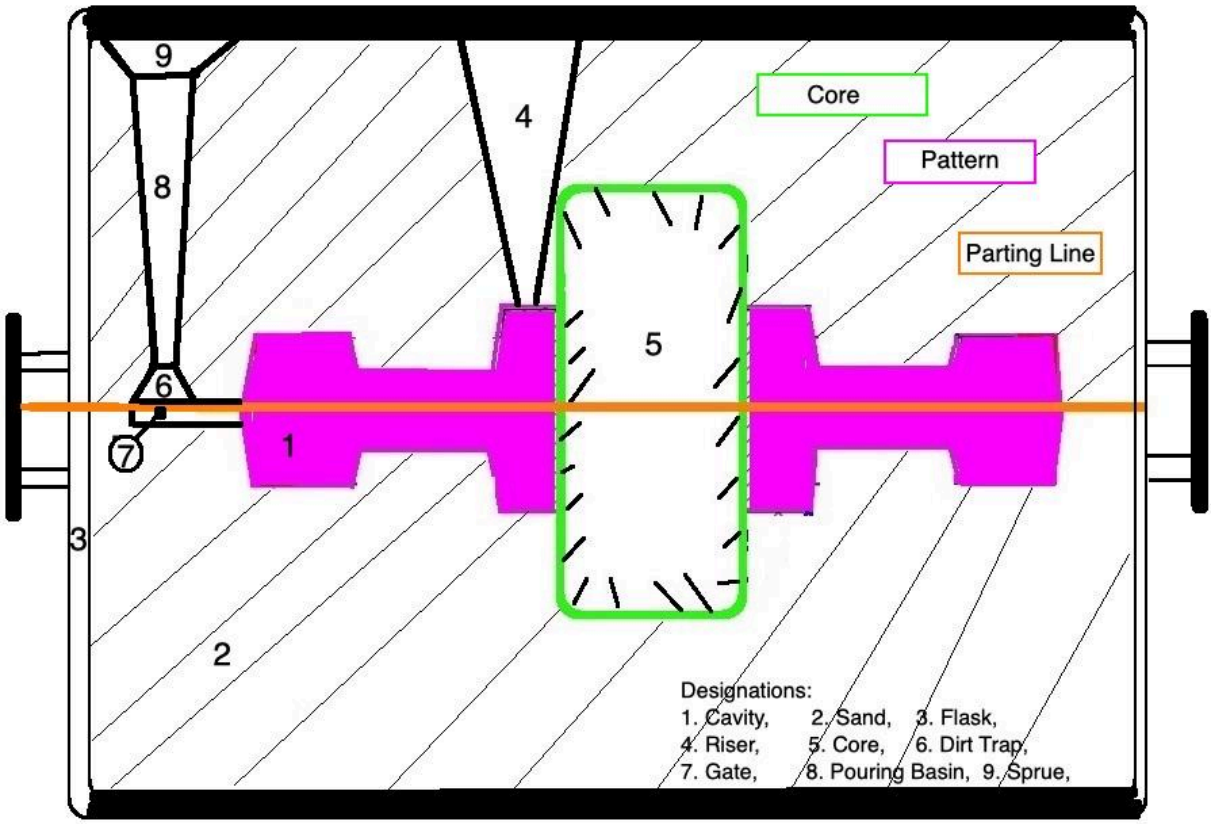


Fig. 6 Casting mould sketch along with mould component designations.

Machining

1. Analysis of part technical drawing - Find the surfaces marked for machining. Mark the machined surfaces according to turning type. What type of machining (operation) and tools will be used to achieve the final shape of the part? (10 p)

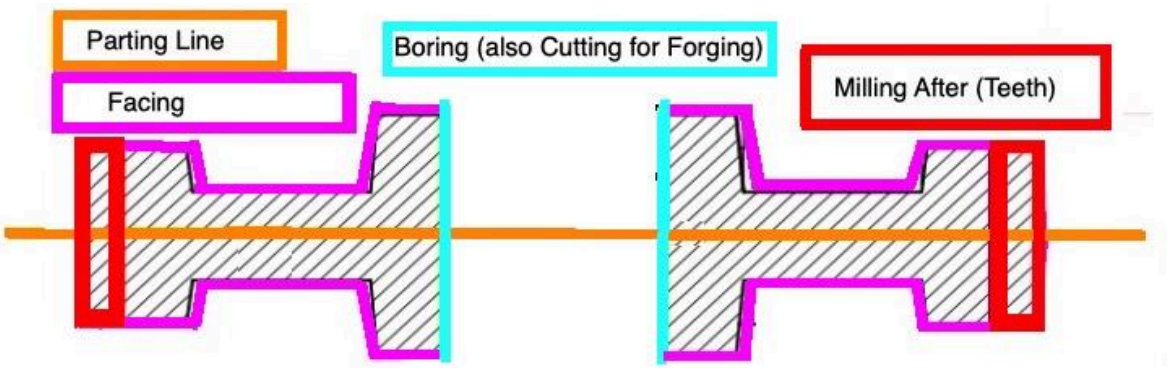


Fig. 7 Machining surfaces and tools.

*The final overall shape of the part will be achieved by a combination of **turning, boring, facing, and milling** operations. The surfaces marked for machining are:*

- ***Facing**: Both ends of the part (left and right faces) will be faced to remove excess material (flash) and to ensure flat, clean surfaces. This is typically done using a **facing tool on a lathe**.*

- ***Boring**: The central hole is finished with boring to refine the diameter and surface finish. In forging, boring also removes the central flash. This is done using a **boring bar on a lathe**.*

- ***Milling (Teeth)**: The external features like **gear teeth or splines** are created using **a milling cutter** on a **milling machine**. This is done after turning and boring.*

***Machining Tools Used**:*

- *Lathe (turning, facing, parting, boring)*

- *Milling machine (for cutting external teeth)*

*In the case of **casting**, the machining steps are similar, but since a core is used, there is no central flash to remove. Only external flashes from gates and runners need to be cut off.*

2. Machining steps - How many fixing steps are necessary to achieve the main shape? (16 p)

Answer:

*To achieve the final shape of the part, the following **fixing steps** (setups) are necessary:*

1. ***First Fixing (Lathe Setup 1)**:*
    - *Face the left side (removal of flash and preparation of a reference surface)*
    - *Bore the internal hole (removal of central flash for forging or finishing for casting)*
2. ***Second Fixing (Lathe Setup 2 - Flip part)**:*
    - *Face the right side (ensure overall length and clean surface)*
    - *Final internal boring if deeper refinement is needed*
3. ***Third Fixing (Milling Setup)**:*
    - *Mount part on milling machine*
    - *Mill external features such as teeth or splines*

Total: 3 Fixing Steps

These steps ensure proper alignment, surface finish, and dimensional accuracy of the final part.